

Martingale Innovations from Recurrent Networks for Sequential Change-Point Detection

YUAN-CHIN IVAN CHANG^a

Institute of Statistical Science, Academia Sinica, 128 Academia Road, Section 2, Nankang, Taipei 115, Taiwan.

^aycchang@sinica.edu.tw

Sequential monitoring is challenging when observations are serially dependent, high-dimensional, and not governed by a trusted parametric model. This paper revisits the innovations principle using a learned predictor: a recurrent network removes the serially predictable component of the data stream, and a CUSUM statistic monitors the resulting prediction innovations. The central goal is to show that the false-alarm and detection-delay guarantees of classical change-point theory extend, essentially intact, to black-box learned predictors. The theoretical development has two parts. First, for contractive recurrent networks, we show that the hidden state admits a Doob decomposition into a predictable component and a martingale-difference innovation. This innovation stabilizes geometrically, and its partial sums satisfy a functional central limit theorem. Second, for any predictor whose conditional residual drift is almost surely bounded and whose residuals have sub-Gaussian tails, we establish an exponential lower bound on the in-control average run length of the Innovation CUSUM, with the optimal exponent uniformly over this class, together with a logarithmic upper bound on detection delay. In the Gaussian mean-shift setting, the proposed statistic reduces to the classical likelihood-ratio CUSUM and is first-order minimax optimal. These guarantees require no parametric model for the observed dynamics, although they are not assumption-free. We therefore provide conditions under which a learning guarantee implies the required drift condition with high probability. We also characterize when learning improves monitoring performance: under a nonlinear conditional mean, linear residualization incurs an irreducible error that a learned predictor can eliminate. Simulations and real-data applications illustrate the theory.

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1. Introduction

Data streams requiring monitoring for structural change are now ubiquitous. Intensive-care units track dozens of physiological signals, public-health agencies monitor infectious-disease incidence, industrial systems record hundreds of correlated sensor measurements, and risk desks follow panels of asset returns. In each setting, a detector must determine online whether the data-generating mechanism has changed. The two types of error carry different costs. A *false alarm*—signaling a change while the system remains in control—may trigger an unnecessary clinical intervention, halt a production line, or prompt premature portfolio rebalancing. A missed or delayed detection, by contrast, may leave a deteriorating patient unattended, an equipment fault unresolved, or a financial loss unhedged. The statistical objective is therefore not simply to detect change, but to control false alarms without incurring excessive detection delay.

This trade-off is particularly difficult in the settings considered here. Observations are serially dependent, the dependence structure is generally unknown and seldom specified with sufficient confidence to support a parametric monitoring procedure, and the number of monitored coordinates may be large. This paper studies sequential change-point detection under the combined challenges of unknown serial dependence, limited model specification, and high dimensionality.

Several established approaches address components of this problem. Classical CUSUM procedures are powerful—and, under standard likelihood-ratio formulations, optimal or asymptotically optimal—when observations are independent and the pre- and post-change distributions are correctly specified (Page, 1954, Lorden, 1971). Offline segmentation methods (Killick, Fearnhead and Eckley, 2012) and Bayesian online change-point procedures (Adams and MacKay, 2007) provide flexible alternatives for identifying structural changes. Their standard theoretical guarantees, however, typically depend on assumptions about the data-generating process and do not automatically extend to settings with unknown serial dependence or increasing dimension. Classical time-series models address dependence explicitly. Autoregressive models represent temporal structure through a specified lag order, hidden Markov models encode dependence through a latent finite-state process, and Kalman filters propagate a finite-dimensional Gaussian state estimate (Box et al., 2015, Hamilton, 1994). These models permit tractable monitoring rules and rigorous guarantees when their structural and distributional assumptions are adequate, but misspecification can distort both false-alarm behavior and detection performance. A less developed objective is therefore to construct a monitoring procedure whose false-alarm guarantees remain valid under weak assumptions on serial dependence while accommodating high- or growing-dimensional observations.

A further strand of the literature monitors *residuals* rather than raw observations. A seminal formulation of real-time monitoring with estimated parameters is given by Chu, Stinchcombe and White (1996), while Bai (1994) establishes weak-convergence results for sequential empirical processes of estimated residuals in ARMA models. Lai (1995) develops a broader theory of sequential change-point detection for quality-control and dynamical-system settings, and Aue and Horváth (2013) survey methods for detecting structural breaks in time series. Much of this literature constructs residuals from a specified parametric model, so its theoretical guarantees depend on the adequacy of that model and its estimation. In a complementary direction, conformal test martingales and e-detectors (Vovk, Gammerman and Shafer, 2005, Shin, Ramdas and Rinaldo, 2024) provide nonparametric, anytime-valid monitoring. Standard conformal constructions rely on exchangeability or related invariance assumptions, whereas e-detectors require valid e-values under the in-control null, often obtained from a known reference distribution or a suitably specified null class. The present paper lies between these two traditions. Like residual-based procedures, it monitors one-step-ahead prediction errors, but the predictor is learned rather than specified parametrically. Like e-detector methods, it does not impose a parametric model on the observed dynamics; instead, its in-control guarantees rely on explicit conditions on the conditional residual drift and tails, and the monitoring statistic is Page’s classical CUSUM.

This paper follows a strategy with a long history in time-series analysis: transform the observations into innovations and then monitor those innovations. The innovations principle states that the one-step-ahead error of the conditional-mean predictor is orthogonal to the observed past—or, probabilistically, forms a martingale-difference sequence—and therefore contains no remaining component that is predictable from that past. This principle underlies Wiener–Kolmogorov prediction theory, the innovations approach to filtering (Kailath, 1968), residual diagnostics in Box–Jenkins modeling (Box et al., 2015), and residual-based methods for detecting abrupt changes (Basseville and Nikiforov, 1993). Under a correctly specified linear-Gaussian state-space model, the principle takes an especially strong form: Kalman-filter innovations are independent Gaussian variables, and a likelihood-based CUSUM constructed from them retains the corresponding classical false-alarm guarantees. Outside this setting, obtaining exact innovations generally requires knowledge of the relevant conditional distribution or predictor—precisely what is unavailable when the dependence structure is unknown. Recurrent neural networks (RNNs) (Rumelhart, Hinton and Williams, 1986), long short-term memory networks (LSTMs) (Hochreiter and Schmidhuber, 1997), gated recurrent units (GRUs) (Cho et al., 2014), and selective state-space models such as Mamba (Gu and Dao, 2023) offer flexible ways to learn predictive state representations from data. Although their architectures still impose a state-update structure, they can learn the effective memory and

nonlinear predictive relationships rather than requiring a prespecified time-series model. They therefore provide a natural modern implementation of the innovations principle: use a learned predictor to remove, as far as possible, the serially predictable component of the stream, and monitor the resulting residuals. The question addressed in this paper is which classical CUSUM guarantees remain valid when the predictor is learned rather than specified in advance.

Two questions must be answered. First, in what precise probabilistic sense does a learned recurrent predictor produce innovations, and how rapidly does its state—and hence the distribution of those innovations—stabilize? This determines whether calibration based on a training segment remains valid during subsequent monitoring. Second, what false-alarm and detection-delay guarantees can be established for a CUSUM applied to the learned innovations, and do those guarantees remain valid as the observation dimension grows?

Our results are deliberately two-tier. First, for a basic RNN whose recurrent weight matrix satisfies an operator-norm contraction condition (Assumption 2.1)—enforceable by spectral normalization (Miyato et al., 2018)—the hidden state admits a Doob decomposition into a predictable component and a martingale-difference innovation. The innovation converges geometrically to its stationary counterpart, and its partial sums satisfy a functional central limit theorem.

Second, the detection theory is architecture-agnostic. Bayes prediction residuals are martingale differences (Proposition 2.3), while fitted residuals are controlled through their conditional mean drift. For contractive RNNs, Proposition 2.4 bounds this drift by separating initialization, approximation, and estimation errors. The CUSUM results then require only bounded conditional drift and conditional sub-Gaussian tails, yielding an in-control average-run-length lower bound (Theorem 3.1), an out-of-control delay upper bound (Theorem 3.3), and the resulting exponential-false-alarm/logarithmic-delay trade-off (Corollary 3.4). Thus, the RNN theory helps verify the residual conditions, while the CUSUM guarantees apply to any predictor satisfying them.

The contributions comprise four connected results. First, under operator-norm contractivity, the recurrent hidden state is uniformly (L^2)-bounded and admits a Doob decomposition whose martingale-difference innovation converges geometrically to stationarity (Theorem 2.1); its partial sums satisfy a functional central limit theorem (Corollary 2.2). Second, Proposition 2.4 transfers this state-level stabilization to a residual approximation bound for recurrent predictors with Lipschitz readouts. Third, under explicit conditional-drift and tail conditions, the Innovation CUSUM satisfies exponential in-control ARL and logarithmic detection-delay bounds (Theorems 3.1 and 3.3). These attain the classical minimax detection rate; in the Gaussian mean-shift case, the procedure is first-order minimax optimal, and the Page CUSUM calibrated to the false-alarm constraint is exactly Lorden-optimal (Theorem 3.5). Fourth, with a nonlinear conditional mean, fixed linear residualization incurs an irreducible innovation defect, whereas a consistent learned predictor eliminates it and thereby improves the guaranteed ARL margin (Proposition 2.5 and Remark 2.4). Empirical studies examine these mechanisms under misspecification, short calibration windows, and equal-ARL comparisons with parametric detectors.

The contribution lies in the synthesis. Its ingredients are classical: Doob decomposition; the martingale-difference property of Bayes prediction residuals; geometric forgetting under contractive stochastic recursions (Doukhan, 1994, Miller and Hardt, 2019); martingale functional central limit theory (Hall and Heyde, 1980, Billingsley, 1999); the (L^2)-projection characterization of the best linear predictor (Proposition 2.5); and classical CUSUM theory, including exponential-supermartingale false-alarm arguments (Page, 1954, Lorden, 1971). The novelty is their combination: using contractive recurrent states to construct stabilized innovations, transferring this stability to fitted prediction residuals, and establishing sequential CUSUM guarantees under explicit residual-drift and tail conditions without specifying a parametric model for the observed dynamics.

This perspective differs from work on recurrent architectures that emphasizes stability, expressivity, or efficient sequence modeling (Miller and Hardt, 2019, Miyato et al., 2018, Gonon and Ortega, 2020,

Gu, Goel and Ré, 2022, Gu and Dao, 2023, Orvieto et al., 2023), as well as neural-tangent-kernel and information-bottleneck analyses of fixed-time representations (Jacot, Gabriel and Hongler, 2018, Tishby and Zaslavsky, 2015, Saxe et al., 2019). Here, the learned state removes predictable serial structure, enabling sequential monitoring under explicit drift, tail, and post-change signal conditions without a parametric model for the dynamics.

Sections 2–3 develop the theory, and Section 4 presents four empirical studies. The Supplementary Material contains eight additional studies, background, and proofs. Table B.1 summarizes the paper’s two tiers: contractive-RNN innovation theory and architecture-agnostic detection guarantees.

2. Methodology

We work on a complete probability space $(\Omega, \mathcal{A}, \mathbb{P})$. Let $(X_t)_{t \geq 1}$ be a strictly stationary ergodic process; by Kolmogorov’s extension theorem we may, and do, extend it to a two-sided strictly stationary ergodic process $(X_t)_{t \in \mathbb{Z}}$. Throughout Sections 2–3 the filtration is the *two-sided* natural filtration $\mathcal{F}_t = \sigma(X_s : s \leq t) \vee \sigma(h_0)$, $t \in \mathbb{Z}$, where the RNN initial state h_0 (introduced below) is square-integrable and independent of $(X_t)_{t \in \mathbb{Z}}$; a deterministic h_0 , as in all our experiments, is a special case. Conditioning on the two-sided past is what makes the stationary innovation process of Section 2.1 exactly stationary and adapted; the detector of Section 3 conditions only on the observable one-sided history, and its guarantees are self-contained there. The tail σ -algebra is $\mathcal{T}_\infty = \bigcap_{t \geq 1} \sigma(X_t, X_{t+1}, \dots)$. Throughout Sections 2–3 we use p to denote the hidden-state dimension and d_{in} for the input dimension. An RNN processes the sequence via $h_t = \phi(W_h h_{t-1} + W_x x_t + b_h)$ (with $h_t \in \mathbb{R}^p$) and an LSTM augments this with a gated cell state; both architectures are defined precisely in Section A of the Supplementary Material. In Section 4 the symbol d refers to the observation (input) dimension of the multivariate DGP, following the change-point literature convention. Throughout, we assume the effective recurrent Lipschitz constant satisfies $L_\phi \|W_h\|_{\text{op}} < 1$ (operator-norm contractivity; see Assumption 2.1 below), which can be encouraged in practice by spectral normalization (Miyato et al., 2018). For one-step prediction of a target Y_{t+1} , the central object is the innovation $e_{t+1} = Y_{t+1} - \mathbb{E}[Y_{t+1} | \mathcal{F}_t]$, whose martingale-difference sequence (MDS) property is stated in Proposition 2.3. A process (X_t) is *absolutely regular* (β -mixing) if its mixing coefficients $\beta(k) = \sup_t \mathbb{E}\{\sup_{B \in \sigma(X_{t+k}, X_{t+k+1}, \dots)} |\mathbb{P}(B | \mathcal{F}_t) - \mathbb{P}(B)|\}$ tend to zero. Such assumptions are used only for empirical motivation and local approximation, not to claim that (h_t) alone is a Markov chain. Full definitions, the reverse martingale convergence theorem, and the i.i.d. Hewitt–Savage zero-one law are collected in Section A of the Supplementary Material.

Remark 2.1 (Non-tail-measurable targets). Parts (d)–(e) of Lemma B.1 (Supplementary Material) show that when the target θ is not tail-measurable, the forward and reverse limits *differ*: the forward limit $\mathbb{E}[\theta | \mathcal{F}_\infty^+]$ may incorporate information from the finite initial segment, while the reverse limit $\mathbb{E}[\theta | \mathcal{T}_\infty]$ discards it. This formalizes *selective forgetting*: the reverse martingale extracts only the structurally persistent (tail-measurable) component of any target.

Assumption 2.1 (Effective operator-norm contractivity). Throughout the paper we assume $L_\phi \|W_h\|_{\text{op}} < 1$, where L_ϕ is the Lipschitz constant of ϕ and $\|W_h\|_{\text{op}} = \sigma_{\max}(W_h)$. Spectral normalization (Miyato et al., 2018) can encourage this condition in practice.

2.1. Martingale Decomposition of RNN Hidden States

Throughout this subsection, we work under Assumption 2.1 ($\rho := L_\phi \|W_h\|_{\text{op}} < 1$). The existence and uniqueness of the stationary causal solution $(H_t^*)_{t \in \mathbb{Z}}$ satisfying $H_t^* = \phi(W_h H_{t-1}^* + W_x X_t + b_h)$,

$H_t^* \in \sigma(X_t, X_{t-1}, \dots)$, and the geometric coupling bound $\|h_t - H_t^*\|_{L^2} \leq \rho^t \|h_0 - H_0^*\|_{L^2}$ are proved directly in Proposition B.2 (Section B.16 of the Supplementary Material). Thus, the chain of results in this paper does not depend on any external or unpublished result. The companion paper Chang (2026) studies the complementary question of hidden-state convergence under the same contraction condition from a reverse-filtration perspective; none of its results are used here. We write $\Delta_0 := \|h_0 - H_0^*\|_{L^2}$, $D_t^* := \mathbb{E}[H_t^* | \mathcal{F}_{t-1}]$ (the stationary predictable component), and $M_t^* := H_t^* - D_t^*$ (the stationary MDS innovation). Here, the two-sided filtration used in the setup is essential: H_t^* is a fixed measurable functional of $(X_s)_{s \leq t}$ and is therefore $\mathcal{F}_t$ -measurable, while D_t^* is given by the same functional of the shifted path at each t (conditioning additionally on the independent $\sigma(h_0)$ leaves it unchanged). Consequently, $(M_t^*)_{t \in \mathbb{Z}}$ is a strictly stationary, ergodic MDS adapted to $(\mathcal{F}_t)$. Under the one-sided filtration $\sigma(X_1, \dots, X_{t-1})$, neither exact stationarity nor adaptedness would hold because the pre-sample past influences H_t^* . The stationary causal solution is generally a *random* process; it should not be identified with either a deterministic mean-field fixed point or a tail projection.

The Doob decomposition in part (ii) of the following theorem is, on its own, elementary: subtracting its conditional expectation turns any square-integrable adapted process into a predictable term plus a martingale difference. The substance of Theorem 2.1 is what the contractive recursion adds to this decomposition — uniform L^2 control, the *geometric stabilization* of the innovation toward its stationary counterpart (iv), and the stationary innovation covariance Σ_M^* (v) — which are the ingredients the functional CLT (Corollary 2.2) and the Innovation CUSUM (Section 3) actually consume.

Theorem 2.1 (MDS Decomposition with Stabilization Rate). *Let $(X_t)_{t \geq 1}$ be a strictly stationary ergodic process with $\mathbb{E} \|X_t\|^2 < \infty$, and let $(h_t)_{t \geq 0}$ be the RNN hidden-state sequence with initial state h_0 as in the setup (square-integrable, independent of the process). Set $\rho = L_\phi \|W_h\|_{\text{op}} < 1$ and $\Delta_0 = \|h_0 - H_0^*\|_{L^2}$. Then, for each $t \geq 2$:*

- (i) (**L^2 -boundedness.**) (h_t) is uniformly L^2 -bounded. When $\phi(0) = 0$ (e.g. tanh, ReLU):

$$\sup_t \|h_t\|_{L^2} \leq \|h_0\|_{L^2} + \frac{L_\phi (\|W_x\|_{\text{op}} \|X_1\|_{L^2} + \|b_h\|)}{1 - \rho} =: C_0 < \infty.$$

For activations with $\phi(0) \neq 0$ (e.g. sigmoid), an additional term $\|\phi(0)\|/(1 - \rho)$ is added to C_0 (from $\|\phi(u)\| \leq \|\phi(0)\| + L_\phi \|u\|$, so the constant enters undamped by L_ϕ); finiteness is unchanged.

- (ii) (**MDS decomposition.**) The Doob decomposition $h_t = D_t + M_t$ holds, where $D_t := \mathbb{E}[h_t | \mathcal{F}_{t-1}]$ is $\mathcal{F}_{t-1}$ -measurable (predictable) and $M_t := h_t - D_t$ satisfies $\mathbb{E}[M_t | \mathcal{F}_{t-1}] = 0$ a.s. The pair $(M_t, \mathcal{F}_t)_{t \geq 2}$ is a martingale difference sequence.
- (iii) (**Innovation L^2 bound.**) Writing $\text{Var}_2(X_t) := \mathbb{E} \|X_t - \mathbb{E} X_t\|^2$,

$$\mathbb{E} \|M_t\|^2 \leq L_\phi^2 \|W_x\|_{\text{op}}^2 \text{Var}_2(X_t) =: \sigma_M^2. \quad (1)$$

- (iv) (**Innovation stabilization rate.**) The MDS innovations converge to their stationary counterparts at a geometric rate:

$$\|M_t - M_t^*\|_{L^2} \leq 2\rho^t \Delta_0. \quad (2)$$

- (v) (**Conditional variance decomposition and stationary innovation covariance.**) The law-of-total-variance decomposition holds exactly:

$$\mathbb{E} \|h_t - \mathbb{E} h_t\|^2 = \mathbb{E} \|D_t - \mathbb{E} h_t\|^2 + \mathbb{E} \|M_t\|^2. \quad (3)$$

In the stationary regime the innovation covariance matrix

$$\Sigma_M^* := \mathbb{E}[M_t^* (M_t^*)^\top] \quad (4)$$

is well-defined, independent of t , positive semi-definite, and satisfies $\Sigma_M^* \preceq \sigma_M^2 I_p$. Moreover $\lambda_{\min}(\Sigma_M^*) > 0$ under the following nondegeneracy condition: for every nonzero $v \in \mathbb{R}^p$,

$$\text{Var}(v^\top \phi(W_h H_{t-1}^* + W_x X_t + b_h) \mid \mathcal{F}_{t-1}) > 0 \quad a.s. \quad (5)$$

Under linear activation (5) holds precisely when $v^\top W_x \text{Var}(X_t \mid \mathcal{F}_{t-1}) W_x^\top v > 0$ a.s. for every $v \neq 0$, i.e. when the conditional covariance of X_t is positive definite on the range of $W_x^\top$. Unpredictability of X_t together with $\text{rank}(W_x) = p$ is not sufficient: for $W_x = I_2$ and $X_t = (\varepsilon_t, 0)^\top$ with ε_t unpredictable, the second innovation coordinate has zero variance. For nonlinear ϕ the condition can also fail for saturating activations or when $d_{\text{in}} < p$ (see Remark 2.3).

Proof. See Section B of the Supplementary Material. $\square$

Remark 2.2. Parts (i)–(iii) extend Proposition 2.3 to the hidden state. **Parts (iv) and (v) are new:** Part (iv) establishes convergence of the transient innovation at rate ρ^t , requiring burn-in $\log(\Delta_0/\varepsilon)/(-\log \rho) \approx \log(\Delta_0/\varepsilon)/(1-\rho)$, while Part (v) gives Σ_M^* for the functional CLT and Mahalanobis CUSUM scoring.

Remark 2.3 (Scope of the nondegeneracy condition). Condition (5) ensures $\lambda_{\min}(\Sigma_M^*) > 0$ by requiring non-zero innovation variance in every direction. For linear activations, full row rank $\text{rank}(W_x) = p$ is sufficient, but it may fail for saturating nonlinearities on nearly flat regions. When $d_{\text{in}} < p$, full row rank is impossible and, under linear or locally linearized activations, $\text{rank}(\Sigma_M^*) \leq d_{\text{in}}$. Nonlinear transformations can exceed this bound, so rank deficiency cannot be inferred from $\text{rank}(W_x)$ alone. In practice, positive definiteness should be checked from pilot data; Corollary 2.2 assumes it.

Corollary 2.2 (Functional CLT for MDS innovations). In addition to the hypotheses of Theorem 2.1, assume $\mathbb{E} \|X_t\|^{2+\delta} < \infty$ for some $\delta > 0$ and that Σ_M^* is positive definite. Define the partial-sum process $\mathbf{S}_T(u) := T^{-1/2} \sum_{t=1}^{\lfloor Tu \rfloor} M_t$, $u \in [0, 1]$. Then

$$\mathbf{S}_T(\cdot) \xrightarrow{d} (\Sigma_M^*)^{1/2} W(\cdot) \quad \text{in } D([0, 1], \mathbb{R}^p), \quad (6)$$

where W is a standard p -dimensional Brownian motion and $D([0, 1], \mathbb{R}^p)$ denotes the Skorokhod space of right-continuous left-limited functions (the natural domain for the step-function partial-sum process). The transient-correction process $\mathbf{T}_T(u) := T^{-1/2} \sum_{t=1}^{\lfloor Tu \rfloor} (M_t - M_t^*)$ satisfies $\sup_{u \in [0, 1]} \|\mathbf{T}_T(u)\| \rightarrow 0$ in probability, with explicit rate $\mathbb{E}[\sup_{u \in [0, 1]} \|\mathbf{T}_T(u)\|] \leq 2\Delta_0\rho/\{\sqrt{T}(1-\rho)\} \rightarrow 0$ (see proof for the sup-norm bound and the functional Slutsky argument), so (6) holds for (M_t) and (M_t^*) simultaneously.

Proof. See Section B of the Supplementary Material. $\square$

Proposition 2.3 (Bayes prediction residuals are martingale differences). Let Y_{t+1} be a square-integrable, $\mathcal{F}_{t+1}$ -measurable prediction target and let $m_t = \mathbb{E}[Y_{t+1} \mid \mathcal{F}_t]$. Then $e_{t+1} = Y_{t+1} - m_t$ is $\mathcal{F}_{t+1}$ -measurable, satisfies $\mathbb{E}[e_{t+1} \mid \mathcal{F}_t] = 0$, and hence is a martingale difference relative to $(\mathcal{F}_t)$. If an $\mathcal{F}_t$ -measurable estimator $\hat{m}_t$ obeys $\sup_t \|\hat{m}_t - m_t\|_{L^2} \leq \eta$, then the fitted residuals $\hat{e}_{t+1} = Y_{t+1} - \hat{m}_t$ are approximate martingale differences in the sense that $\sup_t \|\mathbb{E}[\hat{e}_{t+1} \mid \mathcal{F}_t]\|_{L^2} \leq \eta$.

Proof. See Section B of the Supplementary Material. □

The next proposition makes explicit how the contractive-RNN stabilization result enters the prediction-residual theory. It does not assert that the hidden state is itself the Bayes predictor. Rather, it separates the transient error caused by the initial recurrent state from the approximation and estimation errors of the readout used for prediction.

Proposition 2.4 (Bridge from recurrent states to prediction innovations). *Work under the hypotheses of Theorem 2.1. Let $Y_{t+1} \in L^2$ be a prediction target with Bayes predictor $m_t = \mathbb{E}[Y_{t+1} | \mathcal{F}_t]$. Let $o : \mathbb{R}^p \rightarrow \mathbb{R}^q$ be a readout with Lipschitz constant L_o , and define the recurrent predictor $\hat{m}_t = o(h_t)$. With H_t^* denoting the stationary causal hidden state and $\Delta_0 = \|h_0 - H_0^*\|_{L^2}$,*

$$\|\hat{m}_t - m_t\|_{L^2} \leq L_o \rho^t \Delta_0 + \|o(H_t^*) - m_t\|_{L^2}. \quad (7)$$

Consequently the fitted prediction residual $\hat{e}_{t+1} = Y_{t+1} - \hat{m}_t$ satisfies

$$\|\mathbb{E}[\hat{e}_{t+1} | \mathcal{F}_t]\|_{L^2} \leq L_o \rho^t \Delta_0 + \|o(H_t^*) - m_t\|_{L^2}. \quad (8)$$

If the readout belongs to an estimated class and its stationary readout error obeys $\|o(H_t^*) - m_t\|_{L^2} \leq a_T + \xi_T$, where a_T is approximation error and ξ_T is estimation error, then the residual drift is bounded by $L_o \rho^t \Delta_0 + a_T + \xi_T$.

Proof. See Section B of the Supplementary Material. □

Proposition 2.3 identifies the innovation defect $\|\mathbb{E}[\hat{e}_{t+1} | \mathcal{F}_t]\|_{L^2}$ with the predictor's L^2 bias $\|m_t - \hat{m}_t\|_{L^2}$. The next result uses this identity to explain why a learned predictor is preferable to a fixed parametric residualization when the conditional mean is nonlinear: a linear (e.g. autoregressive) predictor carries an irreducible innovation defect equal to the nonlinearity of m_t , whereas a consistent learned predictor drives the defect to zero.

Proposition 2.5 (Innovation defect and the misspecification gap). *Let (X_t) be strictly stationary with $m_t = \mathbb{E}[X_{t+1} | \mathcal{F}_t] \in L^2$, and let $\hat{m}_t$ be any $\mathcal{F}_t$ -measurable predictor with fitted residual $\hat{e}_{t+1} = X_{t+1} - \hat{m}_t$.*

- (i) **(Defect = bias.)** $\mathbb{E}[\hat{e}_{t+1} | \mathcal{F}_t] = m_t - \hat{m}_t$, so the innovation defect is $\|\mathbb{E}[\hat{e}_{t+1} | \mathcal{F}_t]\|_{L^2} = \|m_t - \hat{m}_t\|_{L^2}$, which vanishes iff $\hat{m}_t = m_t$ a.s.
- (ii) **(Irreducible linear defect.)** Let $L_t \subseteq L^2(\Omega, \mathcal{F}_t, \mathbb{P})$ be the closed subspace of affine functions of the observed history and $m_t^{\text{lin}} = \Pi_{L_t} X_{t+1}$ the best linear predictor. By stationarity the linear-residual defect is the constant $\eta_{\text{lin}} = \text{dist}_{L^2}(m_0, L_0) \geq 0$, and $\eta_{\text{lin}} > 0$ whenever $m_0 \notin L_0$, i.e. the process is nonlinear in mean.
- (iii) **(Consistent learning removes it.)** If $\|\hat{m}_t - m_t\|_{L^2} \leq \eta_T$ with $\eta_T \rightarrow 0$, then the innovation defect $\rightarrow 0$; the learned residuals are asymptotically an exact MDS irrespective of the nonlinearity of m_0 .

Proof. See Section B of the Supplementary Material. □

Remark 2.4 (Consequence for calibration and the ARL guarantee). The ARL guarantee requires $k > \eta$ and improves with $\kappa = k - \eta$ (Theorem 3.1). Under nonlinearity in mean, linear residualization leaves an irreducible L^2 defect $\eta_{\text{lin}} > 0$, whereas a consistent learned predictor removes it. Because

(B1) uses an essential supremum, this L^2 improvement enlarges the guaranteed margin only with the uniform control of Proposition 2.7. Study CUSUM-MISSPEC confirms the gap, with learned innovations markedly whiter than linear residuals.

Proposition 2.5 is qualitative: a consistent learned predictor removes the defect. The next result makes the removal *quantitative*, decomposing the defect exactly into an estimation and an approximation part. It is a *conditional* rate calculation: the rates in part (ii) are assumed, not established here for the recurrent class.

Proposition 2.6 (Rate of the innovation defect; conditional calculation). *Let $\hat{m} = g_{\hat{\phi}}$ be the recurrent predictor fitted by empirical risk minimization over a class Φ from T in-control observations, and write $\eta_T = \sup_t \|\mathbb{E}[\hat{e}_{t+1} | \mathcal{F}_t]\|_{L^2}$ for its innovation defect and $\sigma_e^2 = \mathbb{E}\|X_{t+1} - m_t\|^2$ for the irreducible one-step Bayes risk. Because $\hat{\phi}$ is random, the population risk is taken out-of-sample and conditionally on the training data: $R_{\mathcal{D}_T}(\phi) = \mathbb{E}[\|X'_{t+1} - g_{\phi}(h'_t)\|^2 | \mathcal{D}_T]$, evaluated on an independent stationary trajectory (X'_t, h'_t) (constant in t in the stationary regime); all statements below are conditional on $\mathcal{D}_T$.*

- (i) **(Exact bias–variance split.)** *The squared defect equals the excess prediction risk and splits into estimation and approximation parts,*

$$\eta_T^2 = R(\hat{\phi}) - \sigma_e^2 = \underbrace{\left(R(\hat{\phi}) - \inf_{\phi \in \Phi} R(\phi)\right)}_{\varepsilon_{\text{est}}} + \underbrace{\left(\inf_{\phi \in \Phi} R(\phi) - \sigma_e^2\right)}_{\varepsilon_{\text{app}}},$$

where $\varepsilon_{\text{app}} = \inf_{\phi \in \Phi} \mathbb{E}\|m_t - g_{\phi}(h_t)\|^2$ is the class approximation error for the Bayes predictor.

- (ii) **(Rate, under assumed approximation and estimation bounds.)** *Suppose the width- W class Φ_W approximates the Bayes predictor at rate $\varepsilon_{\text{app}}(\Phi_W) \leq c_A W^{-2s}$ with smoothness index $s > 0$ (the form available for smooth filters approximated by recurrent/reservoir maps, [Gonon and Ortega, 2020](#)), and obeys, for β -mixing in-control data, an excess-risk bound $\mathbb{E} \varepsilon_{\text{est}} \leq c_E W \log T / T$ ([Mohri and Rostamizadeh, 2010](#)). Balancing at $W \asymp (T / \log T)^{1/(2s+1)}$ gives $\mathbb{E} \eta_T = O((T / \log T)^{-s/(2s+1)}) = \tilde{O}(T^{-s/(2s+1)})$. Both bounds are hypotheses: the cited results do not establish these exact rates for the recurrent class used here, whose parameter count grows faster than W and for which mixing enters through an effective sample size.*
- (iii) **(What this does not give.)** *The defect η_T is an L^2 quantity, whereas Assumption 3.1(B1) requires an almost-sure bound on $\mathbb{E}[\hat{e}_t | \mathcal{F}_{t-1}]$. An L^2 rate does not imply an essential-supremum bound, so η_T may not be substituted for η in Theorem 3.1: part (ii) controls a diagnostic, not the constant entering the ARL guarantee. Proposition 2.7 supplies the additional uniform control under which a learning guarantee does verify (B1).*

The next proposition closes that gap, making explicit the price: control must be *uniform* over the reachable state space, not merely in L^2 .

Proposition 2.7 (High-probability verification of the drift condition). *Work under Assumption 2.1 and suppose:*

- (H1) **(Bounded inputs and initial state.)** $\|X_t\| \leq B_X$ and $\|h_0\| \leq B_0$ a.s. (in particular h_0 may be deterministic, as in the experiments); then $\|h_t\| \leq C_h$ a.s. for an explicit constant C_h (see the proof), so h_t and H_t^* lie in the compact set $\mathcal{H} := \{h : \|h\| \leq C_h\}$.
- (H2) **(Uniform approximation.)** *The readout class contains an L_o -Lipschitz $o_{\star}$ with $a_{\infty} := \text{ess sup}_t |o_{\star}(H_t^*) - m_t| < \infty$.*

- (H3) (**Uniform estimation, with high probability.**) The training sample $\mathcal{D}_T$ is independent of the monitored stream (equivalently, $\mathcal{D}_T \subseteq \mathcal{F}_0$, all residual conditions being understood conditionally on $\mathcal{D}_T$), and the fitted readout $\hat{o}$ satisfies, with probability at least $1 - \delta_T$ over $\mathcal{D}_T$, $\sup_{h \in \mathcal{H}} |\hat{o}(h) - o_\star(h)| \leq \varepsilon_T$.
- (H4) (**Conditional tails.**) Conditionally on $\mathcal{D}_T$, the fitted residuals satisfy Assumption 3.1(B2) with parameter σ .

Then, on that training event, $|\mathbb{E}[\hat{e}_{t+1} | \mathcal{F}_t]| \leq a_\infty + \varepsilon_T + L_o \rho^t \Delta_0^\infty$ a.s. for every t , where $\Delta_0^\infty := \text{ess sup} \|h_0 - H_0^*\| \leq B_0 + C_h < \infty$ by (H1). Consequently, for $t \geq t_0$ with $L_o \rho^{t_0} \Delta_0^\infty \leq \varepsilon_T$, Assumption 3.1(B1) holds with $\bar{\eta}_T := a_\infty + 2\varepsilon_T$ from time t_0 onward. Monitoring must therefore be started after the burn-in: let $S_{t_0} := 0$, $S_t := (S_{t-1} + \hat{e}_t - k)^+$ for $t > t_0$, and $\tau_h^{(t_0)} := \inf\{t > t_0 : S_t \geq h\}$. If $k > \bar{\eta}_T$ then, conditionally on $\mathcal{D}_T$, Theorem 3.1 applied from t_0 gives $\mathbb{E}_0[\tau_h^{(t_0)} - t_0 | \mathcal{D}_T] \geq \exp\{2(k - \bar{\eta}_T)h/\sigma^2\}$ on an event of probability at least $1 - \delta_T$.

Proof. See Section B of the Supplementary Material. □

Remark 2.5. (H1)–(H2) are structural, though (H1) restricts attention to bounded inputs, a sub-regime of the L^2 setting used elsewhere and distinct from the Gaussian innovations of Theorem 3.5(ii). (H3) is the substantive price, demanding sup-norm rather than L^2 estimation error. On the compact $\mathcal{H}$ this is the natural regime—uniform convergence over a bounded Lipschitz readout class follows from covering-number arguments under β -mixing—but we do not establish (H3) for a specific architecture, and regard doing so as the main route to making η genuinely estimable.

Proof. See Section B of the Supplementary Material. □

2.2. Extension to Non-Stationary Processes

The results in Section 2.1 require stationarity. We now describe how they extend to the locally stationary setting of Dahlhaus (1997).

Definition 2.1 (Locally stationary process; Dahlhaus 1997). A triangular array $(X_{t,T})_{1 \leq t \leq T}$ is *locally stationary* with transfer function $A^0(u, \cdot)$ if there exists a representation

$$X_{t,T} = \int_{-\pi}^{\pi} e^{i\lambda t} A^0(t/T, \lambda) dZ(\lambda),$$

where $Z(\lambda)$ is a complex-valued orthogonal increment process and $A^0(u, \lambda)$ is a smooth function of the rescaled time $u = t/T \in [0, 1]$.

Corollary 2.8 (Approximate local stationary decomposition). Under the locally stationary model of Definition 2.1, assume that $A^0(u, \lambda)$ is Lipschitz in u uniformly in λ (Dahlhaus, 1997, Assumption 2), and that the array admits an L^2 stationary approximation satisfying $\sup_{t \in B_T(u)} \|X_{t,T} - X_t^{(u)}\|_{L^2} = O(b_T + m_T/T) + o(1)$ (which follows from the Lipschitz condition via Dahlhaus 1997). Fix $u \in (0, 1)$ and let $X_t^{(u)}$ denote the stationary approximating process with transfer function $A^0(u, \cdot)$. Let $h_{t,T}$ and $h_t^{(u)}$ be the corresponding RNN states driven by $X_{t,T}$ and $X_t^{(u)}$, respectively, under the same contractive recurrence and a common square-integrable initialization at the left edge of the local block. Let $B_T(u)$

be any block of at most m_T consecutive time indices satisfying $|t/T - u| \leq b_T$ for all $t \in B_T(u)$. If $b_T \downarrow 0$, $m_T \rightarrow \infty$, and $m_T/T \rightarrow 0$, then

$$\sup_{t \in B_T(u)} \|h_{t,T} - h_t^{(u)}\|_{L^2} = O\left(b_T + \frac{m_T}{T}\right) + o(1). \quad (9)$$

The stationary approximation $h_t^{(u)}$ has the decomposition of Theorem 2.1, with all bounds evaluated under the local law at rescaled time u and spectral density $f(u, \lambda) = |A^0(u, \lambda)|^2$. Thus the martingale-difference decomposition is local and approximate rather than an almost-sure convergence statement to a literal “local tail” limit of the finite triangular array.

Proof. See Section B of the Supplementary Material. □

3. Innovation CUSUM: In-Control False-Alarm Theory

We now establish a rigorous in-control performance guarantee for the Innovation CUSUM, which applies a Page CUSUM to learned prediction innovations. This is the second tier of the theory: exact Bayes residuals form an MDS; fitted residuals are approximately MDS when their conditional drift is small; and this residual-level property yields the alarm-rate bound through a sub-Gaussian exponential-supermartingale argument. No architecture-specific contraction assumption enters the CUSUM theorem; the guarantee applies to any predictor whose residuals satisfy the conditional drift and tail conditions below. Throughout this section, $(\mathcal{F}_t)$ denotes the observable one-sided filtration generated by the monitored data and any training information, relative to which the fitted residuals $\hat{e}_t$ are adapted. The two-sided filtration of Section 2 is not used here.

3.1. Setup and Assumptions

The one-sided Page CUSUM (Page, 1954) is

$$S_0 = 0, \quad S_t = (S_{t-1} + \hat{e}_t - k)^+, \quad \tau_h = \inf\{t \geq 1 : S_t \geq h\}, \quad (10)$$

where $k > 0$ is a reference (slack) value and $h > 0$ is the alarm threshold.

Assumption 3.1 (Innovation CUSUM operating conditions).

- (B1) **(Conditional residual-drift bound.)** In the in-control regime the fitted prediction innovations satisfy

$$\mathbb{E}[\hat{e}_t \mid \mathcal{F}_{t-1}] \leq \eta \quad \text{a.s. for all } t,$$

for some finite constant $\eta \geq 0$. When $\hat{e}_t = Y_t - \hat{m}_{t-1}$ and $m_{t-1} = \mathbb{E}[Y_t \mid \mathcal{F}_{t-1}]$, this condition is equivalent to the uniform bound $m_{t-1} - \hat{m}_{t-1} \leq \eta$ a.s.; the L^2 bounds in Propositions 2.3 and 2.4 should be read as diagnostics or sufficient ingredients under additional uniform-control assumptions, not as a replacement for this conditional drift requirement.

- (B2) **(Conditional sub-Gaussianity.)** In the in-control regime there exists $\sigma > 0$ such that $\mathbb{E}[\exp\{\lambda(\hat{e}_t - \mathbb{E}[\hat{e}_t \mid \mathcal{F}_{t-1}])\} \mid \mathcal{F}_{t-1}] \leq e^{\lambda^2 \sigma^2 / 2}$ for all $\lambda \in \mathbb{R}$. This holds when $\hat{e}_t \in [-B, B]$ a.s. with $\sigma = B$.

- (B3) **(Reference value exceeds conditional drift.)** $k > \eta$; write $\kappa := k - \eta > 0$ for the *net drift margin*.

3.2. Main Result: Exponential ARL Lower Bound

The in-control average run length $\text{ARL}_0 = \mathbb{E}_0[\tau_h]$ is the false-positive metric: a larger ARL_0 means rarer false alarms. Theorem 3.1 below bounds it from below; the complementary false-negative metric — the out-of-control detection delay — is bounded from above in Theorem 3.3 of Section 3.3, and the two combine into the classical exponential-ARL / logarithmic-delay trade-off (Corollary 3.4).

Theorem 3.1 (Sharp in-control ARL lower bound for the Innovation CUSUM). *Under Assumption 3.1, the in-control average run length (ARL_0 ; here $\mathbb{E}_0[\cdot]$ denotes expectation under the in-control distribution, i.e. prior to any change-point) satisfies*

$$\mathbb{E}_0[\tau_h] \geq \exp\left(\frac{2\kappa h}{\sigma^2}\right), \quad (11)$$

where $\kappa = k - \eta$ is the net drift margin and σ^2 the sub-Gaussianity parameter. The exponent $\theta^* = 2\kappa/\sigma^2$ is the sub-Gaussian adjustment exponent of the increment $\hat{e}_t - k$: it is the largest tilt certified by (B1)–(B2), and it coincides with the exact Lundberg coefficient—and hence with the classical i.i.d. CUSUM rate—in the Gaussian benchmark. Equivalently, an in-control ARL of at least L is guaranteed by $h \geq (\sigma^2/2\kappa) \ln L$.

Proof. See Section B of the Supplementary Material. The bound is obtained by a renewal (excursion) argument: at each return of the CUSUM to zero, the conditional probability that the ensuing excursion reaches h is at most $e^{-\theta^* h}$ by optional stopping on the exponential supermartingale $\exp\{\theta^* \sum (\hat{e}_j - k)\}$, so the number of excursions before an alarm stochastically dominates a geometric variable with mean $e^{\theta^* h}$. No union bound is used. $\square$

Remark 3.1 (Sharpness). The ARL grows exponentially in h with rate $2\kappa/\sigma^2$, where $\kappa = k - \eta$ is the margin between the reference value and the in-control conditional drift. The bound weakens as η increases and becomes vacuous as $\eta \rightarrow k$. This exponent is optimal uniformly over the class in Assumption 3.1: Gaussian increments with mean $-\kappa$ and variance σ^2 attain it, matching the exact exponential rate of the classical i.i.d. CUSUM (Siegmund, 1985, Lorden, 1971). Some non-Gaussian laws may have a larger Lundberg root, so the bound is class-sharp rather than pointwise sharp. Its unit prefactor also improves on a Chernoff-plus-union-bound estimate. Thus, $h \geq (\sigma^2/2\kappa) \ln L$ guarantees the target ARL using only η and σ^2 .

Remark 3.2 (Calibration in practice). The constants in Assumption 3.1 are calibrated on a held-out in-control segment, separate from the training and monitored data. The theorem requires a bound on the conditional drift $\mathbb{E}[\hat{e}_t | \mathcal{F}_{t-1}]$, not merely a small unconditional mean. Calibration means, autocorrelation tests, residual-on-past regressions, and the nonlinear predictability checks of Section 4.2 provide diagnostics for η , but not a theorem-valid estimate of the essential supremum in (B1). Likewise, the sub-Gaussian proxy σ must be bounded conservatively; when $\hat{e}_t \in [-B, B]$, one may take $\sigma = B$ as in (B2). Choose k so that $\kappa = k - \eta > 0$, and set h from Theorem 3.1 for the target ARL L . Upper bounds on η and σ are conservative. Section 4 reports the sensitivity of ARL_0 and delay to (k, h) .

Corollary 3.2 (Two-sided Innovation CUSUM). *Let $\tau_h^\pm = \min(\tau_h^+, \tau_h^-)$ be the alarm time of the two-sided CUSUM, where τ_h^+ is the one-sided alarm time of (10) and τ_h^- that of the mirrored recursion $S_t^- = (S_{t-1}^- - \hat{e}_t - k)^+$. If Assumption 3.1 holds for both $\hat{e}_t$ and $-\hat{e}_t$ (in particular, $|\mathbb{E}[\hat{e}_t | \mathcal{F}_{t-1}]| \leq \eta$ a.s. in-control), then $\mathbb{E}_0[\tau_h^\pm] \geq \frac{1}{4} \exp(2\kappa h/\sigma^2)$, with the same exponent $2\kappa/\sigma^2$ as in Theorem 3.1. The empirical studies in Section 4 use this two-sided version.*

Proof. See Section B of the Supplementary Material. A bound of the form $\mathbb{E}_0[\tau_h^\pm] \geq \mathbb{E}_0[\tau_h]/2$ does *not* follow from the one-sided result, since the expectation of a minimum of two stopping times is not controlled by the average of their individual expectations. Instead the per-excursion bound of Theorem 3.1 gives, for each arm, the tail estimate $\mathbb{P}_0(\tau_h^\pm \leq n) \leq 2n e^{-\theta^* h}$ by a union bound over the two arms, and summing $\mathbb{P}_0(\tau_h^\pm > n)$ over $n \leq \lfloor \frac{1}{2} e^{\theta^* h} \rfloor$ yields the stated bound. $\square$

3.3. Out-of-Control Detection Delay

Theorem 3.1 controls the false-positive side. We now bound the false-negative side — the detection delay after a change — completing the sequential-monitoring guarantee. The delay is controlled once the change induces a positive drift in the innovation, the out-of-control counterpart of the in-control conditional-drift condition (B1).

Theorem 3.3 (Out-of-control detection-delay upper bound). *Consider the one-sided Innovation CUSUM (10) with a change-point at time τ . Suppose:*

- (O1) (**Post-change drift.**) *There is $\mu > 0$ with $\mathbb{E}[\hat{e}_t \mid \mathcal{F}_{t-1}] \geq k + \mu$ a.s. for all $t > \tau$;*
- (O2) (**Bounded increments.**) $|\hat{e}_t - k| \leq B$ a.s.

Then the expected detection delay, conditional on no false alarm before the change, satisfies

$$\mathbb{E}[\tau_h - \tau \mid \tau_h > \tau] \leq \frac{h + B}{\mu}. \quad (12)$$

Proof. See Section B of the Supplementary Material. $\square$

Corollary 3.4 (Exponential-ARL / logarithmic-delay trade-off). *Under Assumptions 3.1 and (O1)–(O2), choosing the threshold $h = (\sigma^2/2\kappa) \ln L$ gives simultaneously an in-control guarantee $\mathbb{E}_0[\tau_h] \geq L$ and a detection-delay guarantee $\mathbb{E}[\tau_h - \tau \mid \tau_h > \tau] \leq \mu^{-1} ((\sigma^2/2\kappa) \ln L + B) = O(\ln L/\mu)$.*

Proof. Combine the threshold formula of Theorem 3.1 with (12). $\square$

Remark 3.3 (A model-free guarantee under residual conditions). Corollary 3.4 has the classical sequential-analysis form: false-alarm protection grows exponentially in the threshold, while detection delay grows only logarithmically in the target ARL. The result is model-free with respect to the data-generating dynamics, but it is not distribution-free in the assumption-free sense. It requires the in-control residual-drift and tail conditions of Assumption 3.1 and the post-change signal and boundedness conditions (O1)–(O2). Under those conditions, no parametric model for the observed process is needed. That the resulting $O(\log L)$ delay is in fact order-optimal — and *exactly* minimax optimal in the Gaussian case — is made precise in Theorem 3.5 below.

3.4. Asymptotic Optimality

Corollary 3.4 shows the Innovation CUSUM attains the optimal *order* of detection delay. We now record that it is order-optimal within Lorden’s minimax framework, and is *exactly* minimax optimal in the canonical Gaussian case. For a stopping time τ and change-point $\nu \geq 1$, Lorden’s worst-case detection delay and the false-alarm constraint class are

$$\mathcal{J}(\tau) = \sup_{\nu \geq 1} \text{ess sup } \mathbb{E}_\nu[(\tau - \nu + 1)^+ \mid \mathcal{F}_{\nu-1}], \quad C_\gamma = \{\tau : \mathbb{E}_\infty[\tau] \geq \gamma\},$$

where $\mathbb{E}_\nu$ (resp. $\mathbb{E}_\infty$) denotes the law with change at ν (resp. no change).

Theorem 3.5 (Asymptotic optimality of the Innovation CUSUM). *Set $h = h_\gamma = (\sigma^2/2\kappa)\log \gamma$, so that $\tau_{h_\gamma} \in C_\gamma$ by Theorem 3.1.*

- (i) **(Order optimality.)** *Under Assumption 3.1 and (O1)–(O2), $\mathcal{J}(\tau_{h_\gamma}) \leq (\sigma^2/2\kappa\mu)\log \gamma + B/\mu = O(\log \gamma)$. If, in addition, the conditional log-likelihood ratios satisfy the conditions of the Lorden lower bound (Lorden, 1971) with information number $I \in (0, \infty)$, then $\inf_{\tau \in C_\gamma} \mathcal{J}(\tau) \geq (\log \gamma)/I(1 - o(1))$. Hence τ_{h_γ} is order-optimal. These bounds imply first-order optimality when $2\kappa\mu/\sigma^2 = I$. Assumption 3.1 and (O1)–(O2) alone do not define I .*
- (ii) **(Gaussian innovations.)** *Suppose that the conditional laws of $\hat{e}_t$ given $\mathcal{F}_{t-1}$ are Gaussian with variance σ^2 and means η before the change and $\eta + \delta$ after it, where $\delta > 0$, and set $k = \eta + \delta/2$. Then $\kappa = \mu = \delta/2$ and $2\kappa\mu/\sigma^2 = \delta^2/(2\sigma^2) = I$. Since the conditional log-likelihood-ratio increment is $(\delta/\sigma^2)(\hat{e}_t - k)$, the Innovation CUSUM is a rescaled Page likelihood-ratio CUSUM. Therefore $\mathcal{J}(\tau_{h_\gamma}) \sim (\log \gamma)/I$, so τ_{h_γ} is first-order asymptotically minimax optimal. Exact Lorden optimality (Moustakides, 1986) holds when the Page CUSUM threshold is calibrated so that $\mathbb{E}_\infty[\tau] = \gamma$ exactly. Since h_γ guarantees only $\mathbb{E}_\infty[\tau_{h_\gamma}] \geq \gamma$, exact optimality is not claimed for this analytic threshold.*

Proof. See Section B of the Supplementary Material. □

Remark 3.4 (Interpretation). Part (i) summarizes performance through the quantity $2\kappa\mu/\sigma^2$, which we call the *detection efficiency*; here $\kappa = k - \eta$ and μ are the in-control and post-change margins. Under a Gaussian innovation mean shift of size δ , $\kappa + \mu = \delta$, and the midpoint choice $k = \eta + \delta/2$ maximizes $\kappa\mu$. It yields $2\kappa\mu/\sigma^2 = \delta^2/(2\sigma^2) = I$, the Gaussian information number, so the innovation CUSUM attains the optimal first-order rate.

3.5. Dimension Stability

Corollary 3.6 (Dimension-stable ARL). *Let $\hat{e}_t \in \mathbb{R}^d$ be the multivariate innovation, $\mu_0 = \mathbb{E}[\hat{e}_t]$, $\Sigma_0 = \text{Var}(\hat{e}_t)$, and let $\mathcal{Z}_t := d^{-1}(\hat{e}_t - \mu_0)^\top \Sigma_0^{-1}(\hat{e}_t - \mu_0) - 1$ be the aggregated Mahalanobis score. Write $u_t := \hat{e}_t - \mu_0$ and $b_t := \mathbb{E}[u_t | \mathcal{F}_{t-1}]$, and assume, uniformly in d :*

- (D1) $\|\mathbb{E}[\hat{e}_t | \mathcal{F}_{t-1}]\| \leq \eta_d = O(1)$ a.s.;
- (D2) $\mathbb{E}[(u_t - b_t)(u_t - b_t)^\top | \mathcal{F}_{t-1}] \preceq \Sigma_0$ a.s.;
- (D3) $\lambda_{\min}(\Sigma_0) \geq c_0 > 0$, so $\|\Sigma_0^{-1}\|_{\text{op}} = O(1)$;
- (D4) $\mathcal{Z}_t$ satisfies Assumption 3.1(B2) with $\sigma_{\mathcal{Z},d} = O(1)$ (e.g. if $\hat{e}_t$ is uniformly bounded in d).

Then $\eta'_d := 4\|\Sigma_0^{-1}\|_{\text{op}}\eta_d^2/d = O(d^{-1}) \rightarrow 0$, so for a fixed reference value k the net margin $\kappa_d = k - \eta'_d \rightarrow k > 0$ and, for all sufficiently large d , the ARL bound (11) holds with $\kappa_d \geq k/2$.

Proof. See Section B of the Supplementary Material. □

Remark 3.5 (Reading (D1)–(D2)). (D1) constrains the *whole* innovation vector, not each coordinate separately. (D2) bounds only the conditional *variance* of the centred innovation, not the squared drift $b_t b_t^\top$; it is therefore strictly weaker than $\mathbb{E}[u_t u_t^\top | \mathcal{F}_{t-1}] \preceq \Sigma_0$, which would force $\mathbb{E}[\mathcal{Z}_t | \mathcal{F}_{t-1}] \leq 0$ and remove the residual term this result is about.

Corollary 3.7 (Raw residualization need not be dimension-stable). *Let $X_t = \Phi X_{t-1} + \varepsilon_t$ be a VAR(1) with $\varepsilon_t \stackrel{\text{iid}}{\sim} (0, I_d)$, $\|\Phi\|_{\text{op}} = \rho_{\text{op}} > 0$ and stationary covariance Σ_X , and form the score from the raw observations ($\mu_0 = 0$, $\Sigma_0 = I_d$). Then*

$$\mathbb{E}[\mathcal{Z}_t^{\text{raw}} \mid \mathcal{F}_{t-1}] = \frac{1}{d} \|\Phi X_{t-1}\|^2, \quad (13)$$

the noise term $d^{-1}\text{tr}(I_d) - 1$ cancelling exactly. This bias need not vanish as $d \rightarrow \infty$: for $\Phi = \rho_{\text{op}} I_d$ the coordinates of X_{t-1} are i.i.d. stationary AR(1) variables with variance $(1 - \rho_{\text{op}}^2)^{-1}$, so by the strong law of large numbers $d^{-1} \|\Phi X_{t-1}\|^2 \rightarrow \rho_{\text{op}}^2 / (1 - \rho_{\text{op}}^2) > 0$ a.s. Hence no fixed k guarantees a positive margin unless it is recalibrated to exceed a bias depending on Φ and the stationary law of X_{t-1} —information that raw CUSUM calibration does not use. (For particular Φ the bias does vanish, e.g. $\Phi = \text{diag}(\rho_{\text{op}}, 0, \dots, 0)$; the point is that it need not.)

Proof. See Section B of the Supplementary Material. □

Remark 3.6 (Connection to Study CUSUM-DIM). Corollary 3.6 provides a formal account of the dimension-scaling pattern observed in Study CUSUM-DIM (Section 4.4): at $d = 10$, the raw and whitened CUSUM ARL₀ falls to 47 and 42 while the Innovation CUSUM rises to 365. Equation (13) shows that the serial-dependence bias η_d^{raw} remains nonzero after the $1/d$ normalization for any $\rho_{\text{op}} > 0$, so a fixed reference value k cannot provide a positive drift margin without explicit recalibration using Φ and Σ_X . This is a calibration requirement, not a universal impossibility: if the reference value k could be perfectly matched to the bias, the raw CUSUM would also be valid. The Innovation CUSUM guarantee holds for all sufficiently large d (specifically once $\eta'_d = 4\|\Sigma_0^{-1}\|_{\text{op}}\eta_d^2/d < k$, which holds for $d \geq d_0$ where d_0 depends on η_d and k).

4. Empirical Studies and Discussion

This section presents four studies aligned with the theory. Section 2 provides the contractive-RNN innovation prototype, whereas the LSTM detector relies on the approximate Bayes-residual MDS condition in Assumption 3.1, not on LSTM contraction. The studies first assess this condition through residual-whiteness diagnostics and then evaluate detection. Study MDS tests the MDS property (Theorem 2.1); Study CUSUM-CP compares LSTM-innovation and classical CUSUM detectors at a fixed threshold, with matched-ARL calibration in Study CUSUM-EQARL; Study CUSUM-DIM examines dimension stability (Corollary 3.6) for $d \in \{1, 2, 5, 10\}$; and Study ETF applies the full pipeline to US equity sector returns around the 2020 COVID crash.

Eight supporting studies are reported in the Empirical Supplement (Section D of the Supplementary Material). Study MDS-VAR extends the MDS diagnostic to multivariate VAR(1) processes; Study KAL benchmarks the LSTM against the Kalman filter on clean and noisy state-space models; Study NILE applies the three-way CUSUM comparison to the Nile River annual discharge series (Cobb, 1978); and Study PELT compares the Innovation CUSUM with the offline PELT detector across the dimensional settings examined in Study CUSUM-DIM. Study CUSUM-EQARL repeats the dimension-scaling comparison under matched false-alarm levels and estimation budgets, showing that the fixed-threshold advantage observed in Study CUSUM-DIM is a finite-sample calibration effect rather than a structural one. Study CUSUM-MISSPEC considers a nonlinear, misspecified process in which the learned innovations are genuinely whiter than those obtained through linear whitening and lead to faster detection, as predicted by Proposition 2.5. Study LS-Diag finds that an LSTM forget gate trained under stationarity systematically adapts poorly to smooth nonstationarity, motivating an explicit correction. Study J-Surr

benchmarks four data-driven surrogates for the predictive-decay profile J_t (defined in Section 4.1); only the estimator corrected using the Kullback–Leibler importance estimation procedure (KLIEP) (Sugiyama, Suzuki and Kanamori, 2012) achieves normalized mean-squared error (NMSE) below one under smooth variation, whereas all four deteriorate under abrupt change.

All code is available in `code/` and can be reproduced using Python 3.9, PyTorch 2.2, and R 4.1 by running `python run_all.py`. Studies MDS and CUSUM-CP use 1,000–2,000 independent test sequences per configuration, whereas Study CUSUM-DIM uses 1,000 test sequences and 2,000 in-control sequences.

4.1. Processes, Benchmarks, and Real Data

Three synthetic data-generating processes and one real dataset are used in the main text; a second real series, the Nile River annual discharge, is analyzed in Study NILE of the Supplementary Material. The first synthetic process is the AR(1) model $X_t = \phi X_{t-1} + \varepsilon_t$, $\varepsilon_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$, with $\phi \in \{0.3, 0.7, 0.95\}$. For the initialization transient starting at $h_0 = 0$, the transient predictive-decay profile is $J_t = |\phi|^t \cdot \sigma / \sqrt{1 - \phi^2}$ (geometric decay). The second is a two-state hidden Markov model (HMM) with states $s \in \{0, 1\}$, transition matrix $P = \begin{bmatrix} 0.9 & 0.1 \\ 0.1 & 0.9 \end{bmatrix}$, and emissions $X_t | s_t \sim \mathcal{N}(\pm 1.5, 1)$; the predictive-decay profile J_t is computed numerically via the forward algorithm. The third is a regime-switching autoregression (RS-AR) with AR coefficients $\phi_1 = 0.30$ and $\phi_2 = 0.85$ under the same transition matrix.

The real dataset is a panel of US equity sector exchange-traded fund (ETF) log-returns described in Section 4.5, examined in a univariate configuration ($d = 1$, using the Financial sector ETF XLF) and a three-dimensional configuration ($d = 3$, XLF together with the Technology sector ETF XLK and the Energy sector ETF XLE), with the 2020-02-19 COVID market peak as the known change-point.

4.2. Study MDS: Stationary Case — Prediction-Residual MDS Diagnostic

We train one InstrumentedLSTM (hidden dimension $p = 32$, spectral normalization on W_h , Adam, $N = 1000$ training sequences of length $T = 150$, 40 epochs) on each process and compute one-step-ahead residuals $r_t = X_{t+1} - \hat{y}_t$ on $N_{\text{test}} = 1000$ independent sequences of length $T = 200$. The Ljung–Box statistic (Ljung and Box, 1978) at maximum lag 5 is applied to each sequence; the *MDS pass rate* is the fraction of sequences not rejecting at level 0.05. This is a diagnostic for linear residual autocorrelation, not a proof of the martingale-difference property. For theorem-level calibration one would also examine conditional drift through residual-on-history regressions or nonlinear predictability checks; the Ljung–Box statistic is used here as a simple, reproducible first screen.

Table 1 shows the results. For AR(1) with $\phi \in \{0.3, 0.7\}$ and the HMM, the LSTM pass rate is close to the nominal level $1 - \alpha = 0.95$, consistent with Proposition 2.3 when the fitted predictor is close to the conditional mean. The lower rate for $\phi = 0.95$ (0.386) is expected: a near-unit-root process requires very long sequences to reach its stationary regime; within $T = 200$ steps the LSTM cell state has not fully converged, so the residuals retain residual autocorrelation. The OLS AR(1) baseline confirms that the pattern is process-driven, not a model artifact: OLS residuals for the HMM and RS-AR have notably low pass rates (0.561 and 0.947) because a misspecified linear predictor cannot capture the nonlinear regime structure.

4.3. Study CUSUM-CP: Innovation CUSUM at a Known Change-Point

Study MDS operated entirely under stationarity. A natural next question is whether an online CUSUM built on LSTM innovations can detect an abrupt structural break reliably, and whether it beats the two

Table 1. Study MDS: Ljung–Box MDS pass rate (fraction of 1000 test sequences not rejecting zero autocorrelation at $\alpha = 0.05$, lag ≤ 5). Bold values are within ± 0.05 of the nominal level 0.95.

Process	LSTM pass rate	OLS pass rate	Expected	N_{seq}
AR(1), $\phi = 0.30$	0.948	0.969	0.95	1000
AR(1), $\phi = 0.70$	0.905	0.974	0.95	1000
AR(1), $\phi = 0.95$	0.386	0.953	0.95	1000
HMM	0.953	0.561	0.95	1000
RS-AR	0.814	0.947	0.95	1000

classical alternatives. This empirically examines the prediction of Theorem 3.1: if the LSTM residuals approximate a martingale difference sequence in-control, the ARL lower bound should be empirically visible.

For abrupt change-point AR, we generate 2000 sequences with a single abrupt change in the AR coefficient from $\phi_{\text{before}} = 0.30$ to $\phi_{\text{after}} = 0.85$ at $\tau^* = 100$ ($T = 200$). All detectors use Page’s two-sided CUSUM (Page, 1954, Lorden, 1971) with the same threshold $h = 5$ and reference value $k = 0.5$, estimated from the first 20% of each sequence. Three methods are compared. M1 (Raw) applies CUSUM to the standardized observations $(X_t - \hat{\mu}_0)/\hat{\sigma}_0$, the standard i.i.d.-calibrated detector. M2 (Whitened) applies CUSUM to AR(1)-whitened residuals $(X_t - \hat{\phi}X_{t-1})/\hat{\sigma}_\varepsilon$, where $\hat{\phi}$ is estimated from the burn-in window; this is the best classical fix, as it explicitly removes lag-1 autocorrelation. M3 (the Innovation CUSUM) applies CUSUM to LSTM one-step-ahead prediction residuals, standardized by in-control moments estimated from a held-out stationary calibration set; *no knowledge of the AR structure is given to the method*.

The key issue with M1 is that the Page CUSUM was designed for i.i.d. input. For AR(ϕ) data, successive standardized observations $(X_t - \mu_0)/\sigma_0$ remain correlated with lag-1 autocorrelation $\approx \phi$, causing the CUSUM statistic to accumulate faster than expected. Concretely, the in-control average run length (ARL_0) — the expected number of steps before a false alarm — is only ≈ 119 for M1, versus the nominal i.i.d. value of 285 at $h = 5$. On sequences of length 200, this means M1 fires before τ^* in 60% of cases (Table 2).

Table 2 shows the three-way comparison at $h = 5$, reporting the two error types separately. The distinction matters here because the raw detector’s apparent “detection” is an illusion of the first-passage rule: M1 issues an alarm on virtually every sequence, but 60.3% of those alarms fire *before* the change (false positives), leaving a correct-detection (true-positive) rate of only 39.7%. M2 (whitened, given the correct AR model) raises the correct-detection rate to 77.2%, and the Innovation CUSUM to 87.7% — the highest — while simultaneously holding the lowest false-positive rate (12.3% versus 22.4% and 60.3%). Misses (false negatives) are negligible for all three ($\leq 0.4\%$). The ARL_0 of M3 (≈ 436) matches that of M2 (≈ 440) and both exceed the i.i.d. nominal (285), indicating that in-control LSTM residuals are slightly under-dispersed relative to the i.i.d. calibration assumption, consistent with the theoretical ARL lower bound of Theorem 3.1 holding with slack.

Detection delay must be read with care, because the tabulated mean averages (alarm $-\tau^*$) over *all* alarms, false positives included. For M3, whose alarms are overwhelmingly correct, the +10.1-step mean is a genuine detection delay; for M1 the -16.3 -step “delay” is not a delay at all but a restatement of its false-positive rate — its alarms predominantly precede the change. Read this way, the two error types tell a consistent story: the Innovation CUSUM attains both the lowest false-positive rate and the highest correct-detection rate, so its false-alarm control is not bought by missing changes. A conditional detection-delay analysis (delay given a correct post- τ^* detection), which removes the false-positive contamination entirely, is the natural complement to this table under a matched- ARL_0 calibration. A representative CUSUM trajectory on a single sequence, illustrating that the Innovation CUSUM alarms

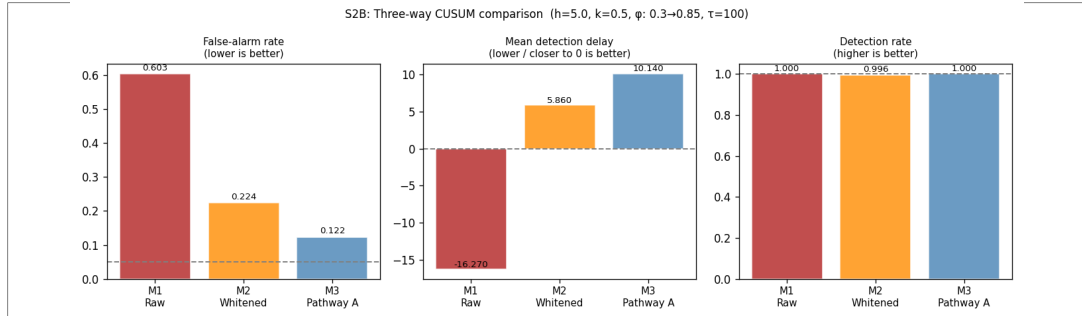


Figure 1. Study CUSUM-CP: three-way CUSUM comparison. Left: false-positive rate (pre- τ^* alarms; dashed line = nominal 5%). Center: mean delay over any alarm (contaminated by false positives for Raw/Whitened; see Table 2). Right: any-alarm rate (*not* correct-detection rate — for Raw this is dominated by pre- τ^* false alarms). The Innovation CUSUM achieves the lowest false-positive rate and, per Table 2, the highest correct-detection rate, without requiring knowledge of the AR model structure.

after the change without being given the AR structure, is shown in Figure D.1 of the Supplementary Material.

Separately, the LSTM’s prediction mean squared error (MSE) more than doubles after the change-point (ratio $\text{MSE}_{\text{post}}/\text{MSE}_{\text{pre}} = 2.25$, versus 1.05 for an oracle model retrained on post-break data), quantifying the cost of model mis-specification after the break. Study J-Surr (Section D.9 of the Supplementary Material) compares four data-driven surrogates for the predictive-decay profile J_t (naive, forget-gate, block-bootstrap, and KLIEP-corrected Sugiyama, Suzuki and Kanamori 2012) and shows that the KLIEP-corrected estimator achieves NMSE below one under smooth non-stationarity, while all four fail under abrupt change.

Table 2. Study CUSUM-CP: three-way CUSUM comparison at $h = 5$. AR(1) change $\phi : 0.30 \rightarrow 0.85$ at $\tau^* = 100$ ($T = 200$, $N = 2000$ sequences). Each replicate is classified by its first alarm into one of three mutually exclusive outcomes: a *false positive* (FP, alarm before τ^*), a *correct detection* (TP, first alarm at or after τ^*), or a *miss* (FN, no alarm in the window); $\text{FP} + \text{TP} + \text{FN} = 1$. Mean delay ‡ is the average of $(\text{alarm} - \tau^*)$ over all replicates that alarm before the final step and therefore *mixes* false positives (negative) with correct detections; it is a clean detection-speed measure only for methods whose FP rate is small. ARL_0 is estimated on an independent in-control population and is the unconfounded false-positive metric. M3 (Innovation CUSUM) uses LSTM prediction residuals; no AR model is provided.

Method	FP (pre- τ^*)	TP (correct)	FN (miss)	Mean delay ‡	ARL_0
M1: Raw CUSUM	0.603	0.397	0.000	−16.3	119
M2: Whitened CUSUM	0.224	0.772	0.004	+5.9	440
M3: Innovation CUSUM	0.123	0.877	0.000	+10.1	436

‡ Averaged over any alarm before the final step, including pre- τ^* false alarms; see text.

MSE degradation (M3 model): pre-change 1.002, post-change 2.252, oracle 1.047.

4.4. Study CUSUM-DIM: Innovation CUSUM Dimension Scaling

The question Study CUSUM-CP leaves open is whether the Innovation CUSUM advantage documented in the scalar setting survives as the observation dimension d grows — the central claim of Corollary 3.6. Study CUSUM-DIM extends the three-way CUSUM comparison to VAR(1) processes with dimension $d \in \{1, 2, 5, 10\}$. The data-generating process is $X_t = \Phi X_{t-1} + \eta_t$ with $\eta_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, I_d)$, where $\Phi = U \text{diag}(\lambda) U^\top$ is constructed so that $\|\Phi\|_{\text{op}} = \max_i |\lambda_i|$. Here and below, ρ_{op} denotes the operator norm of the VAR coefficient matrix, distinct from the RNN contraction constant $\rho = L_\phi \|W_h\|_{\text{op}}$ of Section 2. A change-point at $\tau = 100$ (within a length- $T = 200$ window) raises the operator norm from $\rho_{\text{before}} = 0.30$ to $\rho_{\text{after}} = 0.85$, while U and the noise remain fixed. The three methods are identical to Section 4.3: M1 (raw Mahalanobis CUSUM), M2 (VAR-whitened CUSUM on OLS residuals), and M3 (Innovation CUSUM: LSTM prediction innovations). For $d > 1$ the scalar CUSUM input is the normalized Mahalanobis score $Z_t = d^{-1} \|\Sigma_0^{-1/2} (X_t - \mu_0)\|^2$, centered and standardized; for M3 the LSTM innovation score is analogously aggregated and calibrated on burn-in in-control sequences. Threshold $h = 5$, $N_{\text{test}} = 1000$ change-point sequences, $N_{\text{ARL}} = 2000$ in-control sequences. ARL_0 is the mean alarm time over in-control sequences of length $5T = 1000$, with alarm-free runs right-censored at the sequence length; censoring biases the estimate downward, so the reported ARL_0 values are conservative.

The deterioration of M1 and M2 with dimension has a clear structural explanation, and it is not autocorrelation as such: a correctly centred score with serially correlated increments has no positive *unconditional* drift. The mechanism is the conditional-mean mismatch of (13): the raw score satisfies $\mathbb{E}[Z_t^{\text{raw}} | \mathcal{F}_{t-1}] = d^{-1} \|\Phi X_{t-1}\|^2 > 0$ whenever $\Phi \neq 0$, so the Page CUSUM accumulates a genuine positive conditional drift and ARL_0 falls below its nominal value; serial correlation inflates the variance of the accumulated score and compounds—but does not create—the effect. The problem is compounded in higher dimensions: with d components each contributing correlated mass, the effective autocorrelation of Z_t grows, so ARL_0 collapses as d increases. For M2 (OLS whitening) the same effect applies; moreover, OLS residuals in high-dimensional VAR models carry residual cross-correlation that worsens rather than improves with d . In contrast, the Innovation CUSUM feeds the CUSUM with LSTM *innovations*—the one-step prediction errors of a network trained in-control—which are designed to approximate Bayes prediction residuals. By Proposition 2.3, the exact Bayes residuals are MDSs; when the fitted predictor is close to the conditional mean, the fitted residuals are approximate MDSs. Under this mechanism, the CUSUM threshold $h = 5$ is empirically much better calibrated across d in these experiments.

Table 3 and Figure 2 summarize the results, again separating the two error types. At $d = 1$ all three methods are broadly comparable: $\text{ARL}_0 \approx 150$ –200, false-positive rates of 32%–60%, and correct-detection rates of 40%–68%. As d increases the divergence is dramatic, and it is entirely a false-positive/true-positive story — misses (false negatives) stay below 1% for every method at every d , so nothing is being missed; the question is only whether an alarm, once issued, is correct. At $d = 10$, M1’s false-positive rate reaches 99.9%: it alarms before the change on virtually every replicate, so its correct-detection rate collapses to 0.1% ($\text{ARL}_0 = 47$). M2 is worse still — 100% false positives, 0% correct detections ($\text{ARL}_0 = 42$). The Innovation CUSUM moves in the opposite direction: its correct-detection rate *rises* from 67.9% at $d = 1$ to 83.4% at $d = 10$, its false-positive rate stays in the 16%–32% band, and its ARL_0 grows from 187 to 365. The tabulated mean delays reinforce the reading: M3’s are small and *positive* (genuine post- τ detections), whereas M1 and M2 report -53 and -58 steps at $d = 10$ — not slow detections but alarms firing almost an entire half-window before the change. The lesson is that the “Detect= 1.00” figure one might report for the raw methods is an artifact of the first-passage rule: an alarm is almost always issued, but for M1/M2 at large d it is almost always a false one. The Innovation CUSUM improves the false-positive side (rising ARL_0 , controlled false-positive rate) while *simultaneously* improving the true-positive side, and with misses negligible throughout.

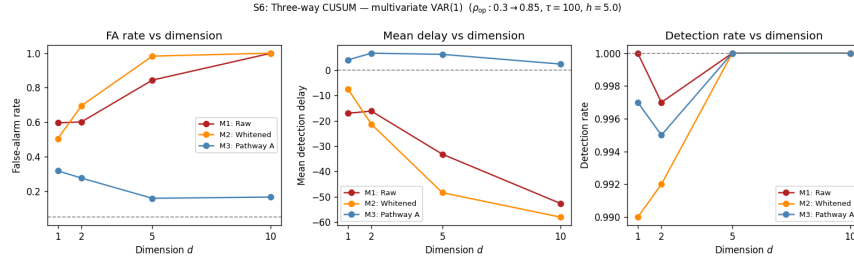


Figure 2. Study CUSUM-DIM: ARL_0 (left) and false-alarm rate (right) as functions of dimension d for the three CUSUM methods. The Innovation CUSUM (green) maintains a high, growing ARL_0 and a stable FA rate, while Raw (blue) and Whitenened (orange) collapse as d increases.

Table 3. Study CUSUM-DIM: multivariate CUSUM comparison ($h = 5, 1000$ test sequences, 2000 ARL_0 sequences). The two error types are reported separately: FP = false-positive rate (first alarm before τ); TP = correct-detection rate (first alarm at or after τ); FN = miss rate (no alarm in the window); $FP + TP + FN = 1$. Delay ‡ = mean of (alarm $- \tau$) over any alarm before the final step (negative = pre- τ); it mixes false positives with correct detections and is a clean speed measure only when FP is small. Bold marks the best TP and ARL_0 per block.

d	Method	FP	TP	FN	Delay ‡	ARL_0
1	Raw (M1)	0.596	0.404	0.000	-17.1	146
	Whitenened (M2)	0.505	0.485	0.010	-7.5	199
	Innovation CUSUM	0.318	0.679	0.003	+4.0	187
2	Raw (M1)	0.602	0.395	0.003	-16.2	145
	Whitenened (M2)	0.696	0.296	0.008	-21.4	126
	Innovation CUSUM	0.276	0.719	0.005	+6.6	231
5	Raw (M1)	0.844	0.156	0.000	-33.2	74
	Whitenened (M2)	0.983	0.017	0.000	-48.4	53
	Innovation CUSUM	0.159	0.841	0.000	+6.2	320
10	Raw (M1)	0.999	0.001	0.000	-52.6	47
	Whitenened (M2)	1.000	0.000	0.000	-58.0	42
	Innovation CUSUM	0.166	0.834	0.000	+2.4	365

‡ Contaminated by pre- τ false alarms for Raw/Whitenened; see text.

We are careful, however, not to over-read this fixed- h comparison. Two features of the setup favor the innovation detector and are not intrinsic to it: at the shared threshold the detectors operate at very different in-control false-alarm levels (ARL_0 from 42 to 365 at $d = 10$), and the classical detectors estimate an $O(d^2)$ nuisance object (Σ_0 , or a VAR matrix) from a short per-sequence in-control window whereas the learned predictor is pretrained on far more data. Study CUSUM-EQARL (Section D.6 of the Supplementary Material) removes both confounds and shows that, at a matched false-alarm level and a matched estimation budget on *these VAR data*, the three detectors are essentially equivalent; the fixed- h collapse is a finite-sample calibration effect, not a structural one. The genuine and provable advantage of learning the predictor appears instead under model misspecification (Proposition 2.5; Study CUSUM-MISSPEC), and in the practically common regime of a short in-control calibration window, where per-sequence estimation of high-dimensional nuisance parameters degrades but a pretrained predictor does not.

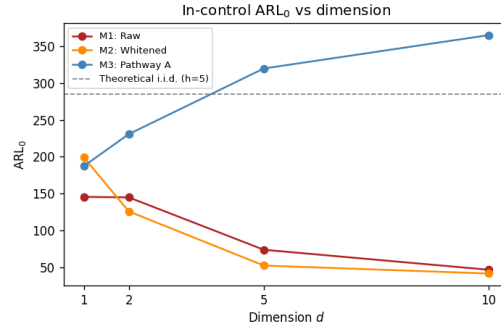


Figure 3. Study CUSUM-DIM: Distribution of detection delay (steps from τ to alarm) across $d \in \{1, 2, 5, 10\}$. The Innovation CUSUM (right column) consistently yields positive delays (valid post-change detection); Raw and Whitenized (left and middle columns) shift to large negative delays as d grows.

4.5. Study ETF: Real Multivariate Application — US Equity Sector Returns

Study ETF addresses the gap that Study CUSUM-DIM leaves open: whether the dimension-stability advantage of the Innovation CUSUM transfers to a real setting where in-control dynamics are non-Gaussian, heteroskedastic, and cross-correlated in a way no parametric VAR fully captures. It should therefore be read as a *stress test* outside the clean theorem assumptions: the sub-Gaussianity of Assumption 3.1(B2) and the conditional-homoskedasticity condition of Corollary 3.6 are known to be violated by heavy-tailed, volatility-clustered equity returns, so the study probes robustness of the qualitative dimension-scaling prediction rather than confirming the bound under its stated hypotheses. Two configurations are examined. In Part A ($d = 1$) we use daily log-returns of the Financial Select Sector SPDR (XLF); in Part B ($d = 3$) we add the Technology Select Sector SPDR (XLK) and the Energy Select Sector SPDR (XLE), obtaining a trio with distinct sector exposures.

Daily log-returns are $r_{t,i} = \log(P_{t,i}/P_{t-1,i})$, sourced from Yahoo Finance. The in-control period is 2017-01-01 to 2019-12-31 ($T_0 \approx 754$ trading days), a sustained low-volatility bull-market regime with stable cross-sector correlations. The test window is 2020-01-01 to 2022-06-30 ($T_1 \approx 630$ days). The known change-point is $\tau^* = 32$ steps into the test window, corresponding to 2020-02-19 (the S&P 500 all-time high immediately before the COVID-19 pandemic crash), after which cross-sector correlations rose sharply and the joint return distribution shifted abruptly (Killick, Fearnhead and Eckley, 2012). The COVID break is preferred over the 2008 financial crisis because the Global Financial Crisis (GFC) involved a prolonged pre-crisis deterioration from mid-2007 onward, making it difficult to identify a single clean change-point; the COVID crash was effectively instantaneous at the cross-sector level.

Because T_0 is large, no bootstrapping is required; the LSTM is trained on rolling windows of length 100 drawn from the in-control period. The threshold and slope parameters are fixed at $h = 5$ and $k = 0.5$, consistent with all other studies. The ARL_0 values in Table 4 are estimated from 500 in-control score sequences of length 500, generated by block bootstrap (block length 10) from the pre-COVID calibration scores; alarm-free runs are right-censored at the sequence length, so the estimates are conservative.

The three CUSUM variants from Study CUSUM-DIM are applied in each configuration. M1 (Raw) forms the Mahalanobis score $Z_t = d^{-1} \|\Sigma_0^{-1/2}(r_t - \mu_0)\|^2 - 1$ using in-control mean μ_0 and Cholesky factor $\Sigma_0^{1/2}$. M2 (Whitenized) first applies a VAR(1) whitening step estimated from the in-control period. M3 (Innovation CUSUM) applies the CUSUM to LSTM one-step-ahead prediction residuals standardized by their in-control moments. Part A ($d = 1$) bridges the univariate results (Study CUSUM-CP, and

Study NILE of the Supplementary Material) and the multivariate synthetic results (Study CUSUM-DIM); Part B ($d = 3$) tests whether the advantage is maintained under genuine cross-sectional dependence.

Table 4 summarizes the results across both configurations ($h = 5$, $k = 0.5$). In Part A ($d = 1$), the Innovation CUSUM alarms at $t = 17$, fifteen steps before the officially designated COVID peak $\tau^* = 32$. Under the formal definition of Section 3.1, any alarm before τ^* constitutes a false alarm. The negative delay should nonetheless be interpreted cautiously: the S&P 500 sell-off began in early February, two weeks before the officially designated February 19 peak, so the true onset of the distributional shift is itself uncertain and the pre- τ^* alarm may reflect genuine early-warning signals in the LSTM innovation process. This illustrates the practical difficulty of separating a false positive from an early true positive when the change onset is itself ill-defined: whether one treats the $t = 17$ alarm as a false alarm or a timely detection depends on the cost asymmetry of the application — in a risk-monitoring setting a two-week-early warning of a market dislocation is typically the desirable error to make, whereas a late alarm (a false negative at the peak) is not. M1 and M2 alarm at $t = 36$ (+4 delay), after the peak. The cleaner, onset-independent comparison is the in-control ARL_0 , estimated entirely within the pre-COVID regime and therefore a pure false-positive-rate measure: M1 and M2 attain 38 and 36 versus the Innovation CUSUM's 43, indicating more frequent false alarms.

In Part B ($d = 3$), the pattern is amplified. M1 and M2 alarm at $t = 24$ (delay -8) while the Innovation CUSUM again alarms at $t = 17$ (delay -15), earlier because the cross-sectional energy (XLE) signal accelerates the Mahalanobis score under the joint distribution shift. Crucially, the Innovation CUSUM's ARL_0 rises from 43 to 67 as d goes from 1 to 3, whereas M1 rises only from 38 to 58 and M2 from 36 to 49. The gap in ARL_0 between the Innovation CUSUM and M1/M2 widens with dimension, consistent with Corollary 3.6 and the simulation evidence in Table 3. Figure 4 displays the CUSUM trajectories for both configurations: in each panel the rapid rise of the Innovation CUSUM statistic in late January–early February 2020 is visible, whereas M1 and M2 remain flat until after the February 19 peak.

Table 4. Study ETF: US equity sector ETF CUSUM results ($h = 5$, $k = 0.5$). Known change-point $\tau^* = 32$ (2020-02-19). ARL_0 estimated from 500 block-bootstrapped in-control score sequences of length 500, right-censored at the sequence length. Delay = alarm time $- \tau^*$ (negative = before the break).

Config	Method	Alarm t	Delay	ARL_0
Part A ($d = 1$, XLF)	Raw (M1)	36	+4	38.1
	Whitened (M2)	36	+4	36.0
	Innovation CUSUM (M3)	17	-15	43.5
Part B ($d = 3$, XLF/XLK/XLE)	Raw (M1)	24	-8	58.1
	Whitened (M2)	24	-8	49.3
	Innovation CUSUM (M3)	17	-15	67.5

4.6. Summary and Interpretation

Taken together, the twelve studies—four in the main text and eight in the Supplementary Material—characterize the finite-sample behavior of the hidden-state innovation, the predictive-decay profile J_t , and the Innovation CUSUM across synthetic and real settings, from univariate autoregressions to non-Gaussian, high-dimensional equity returns. The synthetic studies use many independent replications, enabling statistical comparisons. By contrast, Study ETF and Study NILE (Section D.4) each analyze a single real change-point event and therefore serve as end-to-end demonstrations rather than evidence of a typical real-world advantage.

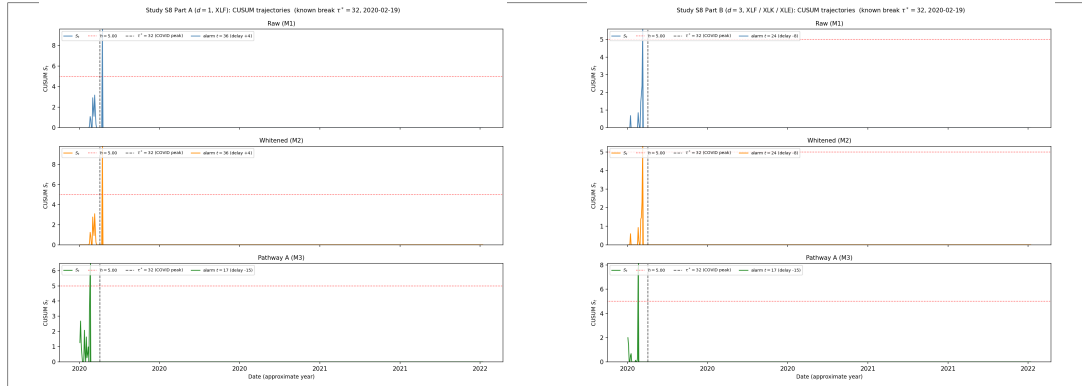


Figure 4. Study ETF: CUSUM trajectories on the COVID test window (2020-01-01 to 2022-06-30); the vertical dashed line marks $\tau^* = 32$ (2020-02-19). *Left* (Part A, $d = 1$, XLF): the Innovation CUSUM (bottom) crosses threshold $h = 5$ at $t = 17$, fifteen steps before the official market peak. *Right* (Part B, $d = 3$, XLF/XLK/XLE): the Innovation CUSUM again alarms at $t = 17$, and its ARL_0 gap over M1/M2 widens relative to Part A, consistent with the dimension-stability prediction of Corollary 3.6.

The dimensional scaling observed at fixed h in Study CUSUM-DIM is striking: raw and whitened ARL_0 collapse as d increases, whereas the Innovation CUSUM’s ARL_0 rises. This result, however, must be interpreted with care. As Study CUSUM-EQARL (Section D.6 of the Supplementary Material) shows, under matched false-alarm levels and estimation budgets, the three detectors are essentially equivalent on these VAR data. The fixed- h collapse arises because the classical detectors operate at a much less conservative point and estimate high-dimensional nuisance parameters from a short in-control window; it does not indicate a structural deficiency. The serial-dependence bias in Corollary 3.6 is real, but once the reference level is calibrated and the nuisance parameters are well estimated, it does not manifest as a dimensional disadvantage in this regime. The advantage of learning the predictor that survives a fully fair comparison is twofold. First, when the conditional mean is nonlinear, fixed linear residualization retains an irreducible innovation defect, whereas the learned predictor does not; this is predicted by Proposition 2.5 and confirmed by the whiteness ordering in Study CUSUM-MISSPEC. Second, in the practically common short-window regime, the learned predictor avoids the ill-conditioned per-sequence estimation that degrades the classical scores. The in-control guarantee is also sharp. The exponent $2(k - \eta)/\sigma^2$ in Theorem 3.1 is the Lundberg coefficient and matches the exact exponential rate of the classical i.i.d. CUSUM (Siegmund, 1985, Ch. 3). Here, that rate is obtained through a renewal argument with no looseness in the exponent.

Study KAL (Section D.3 of the Supplementary Material) shows why high persistence is intrinsically difficult: the analytic decay profile J_t decreases slowly, requiring hidden-state proxies to integrate information over a long history. The LSTM outperforms the EM-Kalman estimator at moderate persistence, but the Kalman estimator regains its advantage when $\phi \approx 0.95$. Study J-Surr (Section D.9 of the Supplementary Material) shows that KLIEP-type corrections achieve NMSE below one under smooth nonstationarity but deteriorate under abrupt changes, thereby delineating when density-ratio reweighting can substitute for retraining.

5. Conclusion

Two practical gaps motivated this work. First, although recurrent networks are widely used for sequential prediction, existing results on their stability and expressivity do not provide the narrower characterization

needed here: a martingale-innovation account of the recurrent state that supports calibration of downstream inference under explicit residual conditions, without requiring a complete account of what the hidden state encodes. Second, classical change-point detectors are typically calibrated for independent observations or fully specified parametric models, so their false-alarm guarantees can fail under unknown serial dependence, especially as the observation dimension grows.

To address the first gap, Theorem 2.1 establishes the Doob decomposition $h_t = D_t + M_t$, together with an explicit geometric L^2 innovation bound and stationary covariance. Corollary 2.2 establishes a functional central limit theorem for the accumulated innovations. To address the second gap, Theorems 3.1 and 3.3 provide a sharp in-control ARL lower bound at the Lundberg rate and a matching detection-delay upper bound. Together, they yield the exponential-false-alarm/logarithmic-delay trade-off in Corollary 3.4 for a detector that imposes no parametric model on the observation dynamics. Theorem 3.5 shows that this trade-off is order-optimal under Lorden’s minimax criterion and first-order minimax optimal under a Gaussian innovation shift; it is exact once the threshold is calibrated to the false-alarm constraint. Propositions 2.4 and 2.5 link state stabilization to residual drift and characterize when a learned predictor achieves a larger guaranteed drift margin—and hence a larger guaranteed ARL exponent—than a fixed linear filter. Corollary 3.6 extends the analysis to the multivariate score. Empirical studies of synthetic autoregressive and vector-autoregressive processes, a nonlinear misspecified process, the Nile discharge series, and US equity-sector returns around the 2020 COVID crash illustrate these mechanisms.

What appears to be new is the *combination* of these ingredients: to our knowledge, no prior work combines recurrent-state innovation theory with sequential CUSUM calibration to provide model-free monitoring guarantees for a detector built on a learned predictor.

Several questions remain open. The apparent dimension-scaling advantage at a fixed threshold is, under a fully matched comparison, a finite-sample calibration effect rather than a structural one (Study CUSUM-EQARL). Characterizing when short-window nuisance estimation favors the learned score therefore remains open, as does developing a fully nonstationary theory beyond the locally stationary approximation of Corollary 2.8. The in-control exponent is already sharp (Theorem 3.1); the remaining slack lies in the constant of the delay bound, which could potentially be sharpened using the overshoot distribution of the reflected random walk. Another natural extension is to relax the sub-Gaussianity and conditional-homoskedasticity assumptions of Corollary 3.6 to accommodate heavy-tailed or heteroskedastic innovations, with direct relevance to finance.

More broadly, the question of what a learned sequential state encodes probabilistically extends beyond RNNs to transformer attention states, diffusion time-series latents, and language-model context representations. Extending the martingale characterization to such representations could place inference and monitoring for these models on the same rigorous footing.

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